



Degradable polyurethane based on triblock polyols composed of polypropylene glycol and ϵ -caprolactone for marine antifouling applications

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Abstract Marine antifouling coatings require superior mechanical properties and effective antifouling. Degradable polyurethane (PU) coatings have a reputation for high performance. Degradable PU films were prepared by mixing triblock polyols [composed of polyether (polypropylene glycol or polyethylene glycol) and ϵ -caprolactone] and crosslinker agent (tolylene diisocyanate trimer). The hydrolytic degradation and water absorption experiments of PU films demonstrated that the PU films could erode in artificial sea water, which were mainly controlled by the hydrophilic property of the polyether. Marine field test revealed that degradable polyurethane had excellent efficiency in preventing biofouling for more than 9 months.

Keywords Triblock polyols, Polypropylene glycol, ϵ -Caprolactone, Degradable polyurethane coating, Marine antifouling

Introduction

Marine biofouling, which is a serious problem for maritime industries and other marine activities, happens due to the undesirable accumulation of microorganisms, plants, and marine animals on submerged surfaces.^{1–5} On ships, biofouling increases hull roughness and drag resistance; in addition, biofouling reduces maneuverability and promotes corrosion.⁶ Marine antifouling coatings are the most reliable way

to prevent marine biofouling.⁷ Broadly speaking, self-polishing acrylate antifouling coatings based on cuprous oxide are still regarded as the most efficient solution for marine antifouling.⁸ Nevertheless, the self-polishing acrylate antifouling coatings show poor resistance to cracking and peeling.⁹ However, good adhesion to underlying coats is an essential requirement for any paint system.¹⁰ Therefore, in order to secure a proper adhesion between the self-polishing coatings and the epoxy undercoat, even after relatively long recoating intervals, tie coat (link coat) usually precedes the antifouling system. At the same time, the properties of traditional acrylate antifouling coatings depend on the sailing speed of a ship, and the coatings exhibit poor properties of antifouling in static vessels and offshore enterprises.^{1–4}

Furthermore, fouling-release (FR) coatings are an attempt to prevent the adhesion of fouling organisms by providing a low-friction, ultra-smooth surface on which fouling organisms have great difficulties in settling.¹¹ At present, FR coatings are limited to fast moving vessels. In addition, FR coatings exhibit poor mechanical properties (poor adhesion to the substrate and are easily damaged).^{12,13}

To meet these demanding requirements, marine antifouling coatings need superior mechanical properties and effective antifouling. The two-package PU coatings have a reputation for high performance.^{14,15} Degradable PU films are attracting growing interest in marine antifouling.^{16–18} The degradable property of PU film (e.g., polycaprolactone-PU) is greatly dependent on the chemistry of its soft segment. It was reported that polycaprolactone-PU (PCL-PU) having a PCL soft segment with a number average molecular weight of over 2000 could crystallize, and its crystallinity increased with increasing number average molecular weight of PCL.^{19,20} The crystallized PU film is not suitable for marine antifouling. Therefore, PU film based on triblock polyols is an approved polymer

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and could be applied for marine antifouling.^{21–23} Polyethylene glycol (PEG) is a nonionic and biocompatible material with outstanding physicochemical properties.²³ However, PEG may not be used for marine antifouling because of extreme hydrophilicity. Nevertheless, polypropylene glycol (PPG), which is hydrophobic, is derived from propylene glycol (PG) monomers with a methyl substituent at the ethylene glycol (EG) basic structure.²⁴ Triblock polyols based on PPG and PCL exhibit mechanical properties that are not shown by the individual homopolymers. However, reports on the application of triblock polyols based on PPG and CL, particularly two-component polyurethane antifouling coatings, have been scarce.

In the present study, a series of triblock polyols were synthesized by a polycondensation reaction using polyether (PPG or PEG) and CL. The degradable PU films were prepared by mixing triblock polyols and a crosslinker agent (tolylene diisocyanate trimer). The hydrophilic property of the polyether chain and the number average molecular weight of the PCL chain were especially investigated. The properties of polyurethane films based on triblock polyols and their use for marine antifouling application are reported in this paper.

Materials and methods

Materials

Polypropylene glycol (PPG₄₀₀, PPG₆₀₀, PPG₈₀₀, PPG₁₀₀₀, and PPG₁₂₀₀) and polyethylene glycol

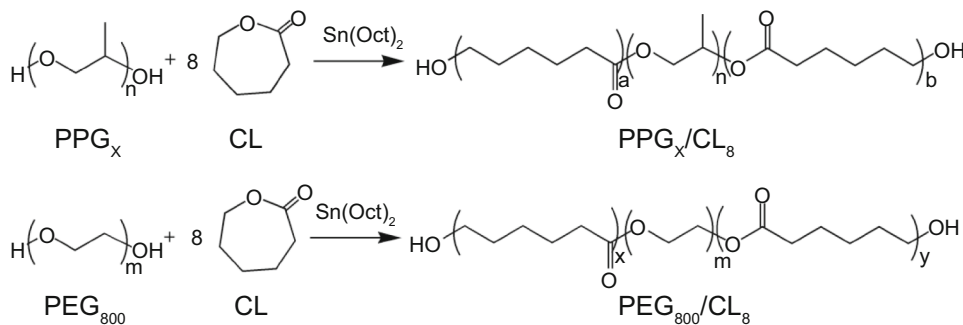
(PEG₈₀₀) from Xiya Chemical Reagent Co., Ltd. were used as received without further purification. ϵ -Caprolactone (CL) (Aladdin) was distilled over CaH₂ before use. Stannous octoate [Sn(Oct)₂] was from Sinopharm Chemical Reagent Co., Ltd. Tolyene diisocyanate trimer (L75) was from Bayer Materials Science and Technology (China) Co., Ltd. Artificial sea water (ASW) was prepared by mixing NaCl (23 g/L), MgCl₂·6H₂O (9.8 g/L), Na₂SO₄·7H₂O (8.9 g/L), and CaCl₂ (1.2 g/L). 4,5-Dichloro-2-octylisothiazolone (DCOIT) and butyl acetate were of industry grade. All other reagents were of analytical grade without further purification.

Synthesis of triblock polyols

The triblock polyols (CL_a-PPG_x-CL_b or CL_m-PEG₈₀₀-CL_n) were synthesized by ring-opening polycondensation reaction using CL and initiator (PPG_x or PEG₈₀₀) as shown in Table 1 and Scheme 1. The synthesis reactions were carried out in a flask immersed in a controlled temperature oil bath. In each polymerization, predetermined amounts of initiator (PPG_x or PEG₈₀₀) and CL at the chosen mass feed ratio shown in Table 1 were simultaneously added. Then, the catalyst Sn(Oct)₂ was added at 60°C and the stirrer was maintained at 150 rpm. The mixture was heated up to 130°C under a dry nitrogen atmosphere until the mixture reached an acid number below 5 mg KOH/g resin. The triblock polyols were named as PPG₄₀₀/CL₈, PPG₆₀₀/CL₈, PPG₈₀₀/CL₈, PPG₁₀₀₀/CL₈, PPG₁₂₀₀/CL₈, and PEG₈₀₀/CL₈ in reference to the components used.

Table 1: Synthesis of triblock polyols

Samples	M_n of PPG _x /n(mol)	PEG ₈₀₀ (mol)	CL (mol)	M_n	M_w/M_n
PPG ₄₀₀ /CL ₈	400/0.1	—	0.8	1432	1.334
PPG ₆₀₀ /CL ₈	600/0.1	—	0.8	1770	1.312
PPG ₈₀₀ /CL ₈	800/0.1	—	0.8	1913	1.292
PPG ₁₀₀₀ /CL ₈	1000/0.1	—	0.8	2157	1.280
PPG ₁₂₀₀ /CL ₈	1200/0.1	—	0.8	2336	1.274
PEG ₈₀₀ /CL ₈	—	0.1	0.8	1836	1.163



Scheme 1: The synthesis scheme of triblock polyols

Preparation of PU films

The triblock polyols were uniformly dissolved in xylene with crosslinker agent (L75) at the ratio of functional groups ($[-NCO]:[-OH] = 1.1:1$). The resulting solutions were poured onto polytetrafluoroethylene plates. The solvent was allowed to evaporate overnight, followed by vacuum drying at 40°C. The dry thickness of PU films was about 200 μm . The PU films were named as PPG₄₀₀/CL₈-PU, PPG₆₀₀/CL₈-PU, PPG₈₀₀/CL₈-PU, PPG₁₀₀₀/CL₈-PU, PPG₁₂₀₀/CL₈-PU, and PEG₈₀₀/CL₈-PU in reference to the triblock polyols used.

Characterization of triblock polyols

¹H NMR spectra of the prepared triblock polyols dissolved in CDCl₃ (10 ~ 15 mg/mL) using 64 scans for proton spectra were acquired on a Bruker V-400M model using tetramethylsilane (TMS) as an internal standard. Chemical shifts (δ) were given in ppm with respect to TMS.

The prepared triblock polyols were characterized for molecular weight using gel-permeation chromatography (GPC). The GPC experiments were performed using PL-GPC 50 Plus system (Agilent Technologies) consisting of refractive index detector and dual angle LS detector, and a set of PLgel columns. The triblock polyols were dissolved in HPLC grade tetrahydrofuran (THF) (1–1.5 mg/mL). The flow rate of carrier solvent (THF) was 1.00 mL/min at 39°C. Polystyrene of a known molecular weight was used as the calibration standard.

FTIR spectra of the prepared triblock polyols were recorded in transmission mode on KBr pellets with a Fisher Nicolet iS 10 IR instrument. The frequency range was from 4000 to 500 cm^{-1} with 32 scans at a resolution of 4 cm^{-1} .

Characterization of polyurethane films

FTIR spectra of PU films were directly recorded in ATR mode with a Fisher Nicolet iS 10 IR instrument.

The T_g values of PU films were measured using DSC instruments (DSC 1, Mettler). Samples (5–10 mg) were heated in the first run to 170°C, then cooled to –70°C and heated in a second scan to 170°C at 10°C/min. The T_g values of PU films were determined from the second heating scan by the half C_p extrapolated tangent method.

X-ray diffraction (XRD) measurements were carried out with a PANalytical instrument. The 2 θ scanning rate was 4°/min, with a scanning range of 10–30°.

Mechanical properties of PU films including tensile strength, tensile modulus, and elongation at break were determined at a strain rate of 15 mm/min. The measurements of PU films were performed at 25°C and stamped out with an ASTM D-638.

Square specimens with dimensions of approximately 20 mm $\times$ 20 mm $\times$ 0.2 mm were then cut from the PU films for water uptake and hydrolytic degradation studies. The various specimens were weighed and subjected to degradation at 25°C in ASW. The samples were removed from the ASW at specified time intervals (1, 2, 3, 5, 10, 15, 20, 30, 40, 50, and 60 days) and washed with deionized water. After wiping, the specimens were weighed and vacuum dried at 60°C.

$$\text{Water uptake (\%)} = \frac{w_2 - w_0}{w_0} \times 100 \quad (1)$$

$$\text{Weight loss (\%)} = \frac{w_1 - w_0}{w_1} \times 100 \quad (2)$$

where w_2 represents the wet weight after wiping, w_0 the dry weight after vacuum drying, and w_1 the initial weight of the specimens. Triplicate specimens were used for each data point.

Characterization of polyurethane coatings

The formulations of the PU antifouling coatings were made by mixing triblock polyols (34 parts), DCOIT (30 parts), xylene (30 parts), and butyl acetate (6 parts). The PU antifouling paint coatings were processed in a paint conditioner for about 8 h. The PU paint coatings were uniformly dissolved with crosslinker agent (L75) at the ratio of functional groups ($[-NCO]:[-OH] = 1.1:1$).

The steel panels (coated with anticorrosion paints) were directly coated with separate PPG_x/CL₈-PU coatings without other tie paints. The adhesion of PU antifouling coatings (about 300 μm) was measured using a pull-off adhesion tester. Pull-off adhesion data of PU antifouling coatings were obtained by detaching an aluminum dolly at a speed of 0.2 MPa/s. Five different points on each sample were tested to obtain an average value.²⁵

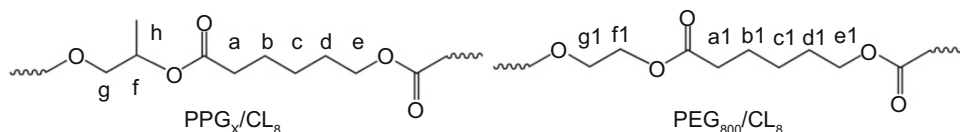
The impact strength of PU coatings was determined by QCJ impact instrument (Shanghai Leao Testing Equipment, China) according to China National Standard (GB/T 1732-93). PU coatings were formed on tinplate substrates. Then, the metal plates were tested at Dart Drop Test Equipment (1000 $\pm$ 1 g in weight, 50 $\pm$ 0.1 cm in height).²⁶

The method for determination of DCOIT release rates of the PU antifouling coatings was based on the “rotating cylinder method.” The test cylinders (100 cm^2) coated with PU antifouling paints were immersed in flowing ASW in a holding tank at 25°C. At specified time intervals (5, 10, 15, 20, 25, 30, 35, and 40 days), the test cylinders were removed from the holding tank and exposed in measuring containers containing 1500 mL of ASW. The test cylinders in measuring containers were rotated at a fixed speed of 60 rpm and 25°C for 1 h, and then 100 mL of ASW was

taken from 1500 mL of ASW, and it was extracted with 10 mL of hexane. The concentration of DCOIT was investigated by an Agilent Cary 60 UV–Vis spectrophotometer at a wavelength of 290 nm.

$$A = \frac{c \times 1500 \times t \times 24}{S} \quad (3)$$

where A is the release rate of DCOIT ($\mu\text{g}/\text{cm}^2/\text{d}$), c is the DCOIT concentration based on the “rotating cylinder method” ($\mu\text{g}/\text{mL}$), t is the immersion time in the individual measuring container (1 h), and s is the surface area coated with PU antifouling paints (100 cm^2).



^1H NMR spectroscopy data of PPG_x/CL_8 (ppm): a (δ 2.25 ~ 2.38), $-\text{CH}_2\text{COO}-$ (PCL); b and d (δ 1.59–1.74), $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ (PCL); c (δ 1.32–1.46), $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ (PCL); e (δ 4.03–4.14), $-\text{CH}_2-\text{O}-$ (PCL); f and g (δ 3.20–3.73), $-\text{CH}_2-\text{CH}-$ (PPG_x); h (δ 1.08–1.20), $-\text{CH}_3$ (PPG_x).

^1H NMR spectroscopy data of $\text{PEG}_{800}/\text{CL}_8$ (ppm): a1 (δ 2.25–2.39), $-\text{CH}_2\text{COO}-$ (PCL); b1 and d1 (δ 1.56–1.74), $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ (PCL); c1 (δ 1.32–1.50), $-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2-$ (PCL); e1 (δ 3.98–4.11), $-\text{CH}_2-\text{O}-$ (PCL); f1 and g1 (δ 3.58–3.73), $-\text{CH}_2-\text{CH}_2-$ (PEG_{800}); from ^1H NMR spectra of triblock polyols, $\text{Sn}(\text{Oct})_2$ -catalyzed ring-opening polymerization of CL was successful in producing triblock polyols.

The steel panels coated with the PU antifouling paints were immersed in ASW and then were taken out at specified time intervals (1, 2, 3, 4, 5, 6, 7, and 8 weeks), and the panels were cleaned between two field test periods. The thicknesses of PU coatings were tested by a coating thickness gauge (Surfix, Germany).²³

The marine field tests for the marine antifouling performances of PPG_x/CL_8 -PU coatings were performed by static raft testing from November 2017 to July 2018 (9 months) at the inner Xiamen Bay ($24^\circ 45' \text{ N}$, $118^\circ 07' \text{ E}$) in China, where the marine biofouling is heavy because of the rich fouling organisms (e.g., balanus, oyster, mussel, bryozoan, and alga). The steel panels were coated with anticorrosion paints. Subsequently, the steel panels were directly coated with separate PPG_x/CL_8 -PU coatings without other tie paints. The steel panels were lowered into sea water at the depth of 0.2–1.0 m. At specified time intervals (3, 6, and 9 months), the steel panels were taken out of the sea and photographed.²²

Results and discussion

Properties of triblock polyols

^1H NMR spectroscopy of triblock polyols

The chemical structures of triblock polyols were confirmed through ^1H NMR spectra (Fig. 1).

Average molecular weights of triblock polyols

The number average molecular weights and molecular weight distributions of triblock polyols are summarized in Table 1. The average molecular weight improved with increasing the number average molecular weights of polypropylene glycol, as shown in Table 1.

FTIR spectra of triblock polyols

The FTIR spectra of triblock polyols are shown in Fig. 2. The characteristic band of $-\text{OH}$ stretching vibrations is at 3492 cm^{-1} . The characteristic band of $-\text{CH}_3$ of PPG_x/CL_8 is at 2971 cm^{-1} . The absorption bands at 2931 and 2861 cm^{-1} are due to asymmetric and symmetric stretching of $-\text{CH}_2-$, and it can be seen that the characteristic bands of $-\text{CH}_2-$ include the band at 1459 cm^{-1} . The stretching absorption of carbonyl group is at 1735 cm^{-1} . The characteristic bands of $\text{C}-\text{O}-\text{C}$ are at 1249 and 1109 cm^{-1} . The ^1H NMR spectra, GPC, and FTIR analysis thus confirmed that triblock polyols were prepared successfully.

Properties of PU films

ATR-FTIR spectra of PU films

The ATR-FTIR spectra of PU films are shown in Fig. 3. The stretching absorption of amide group

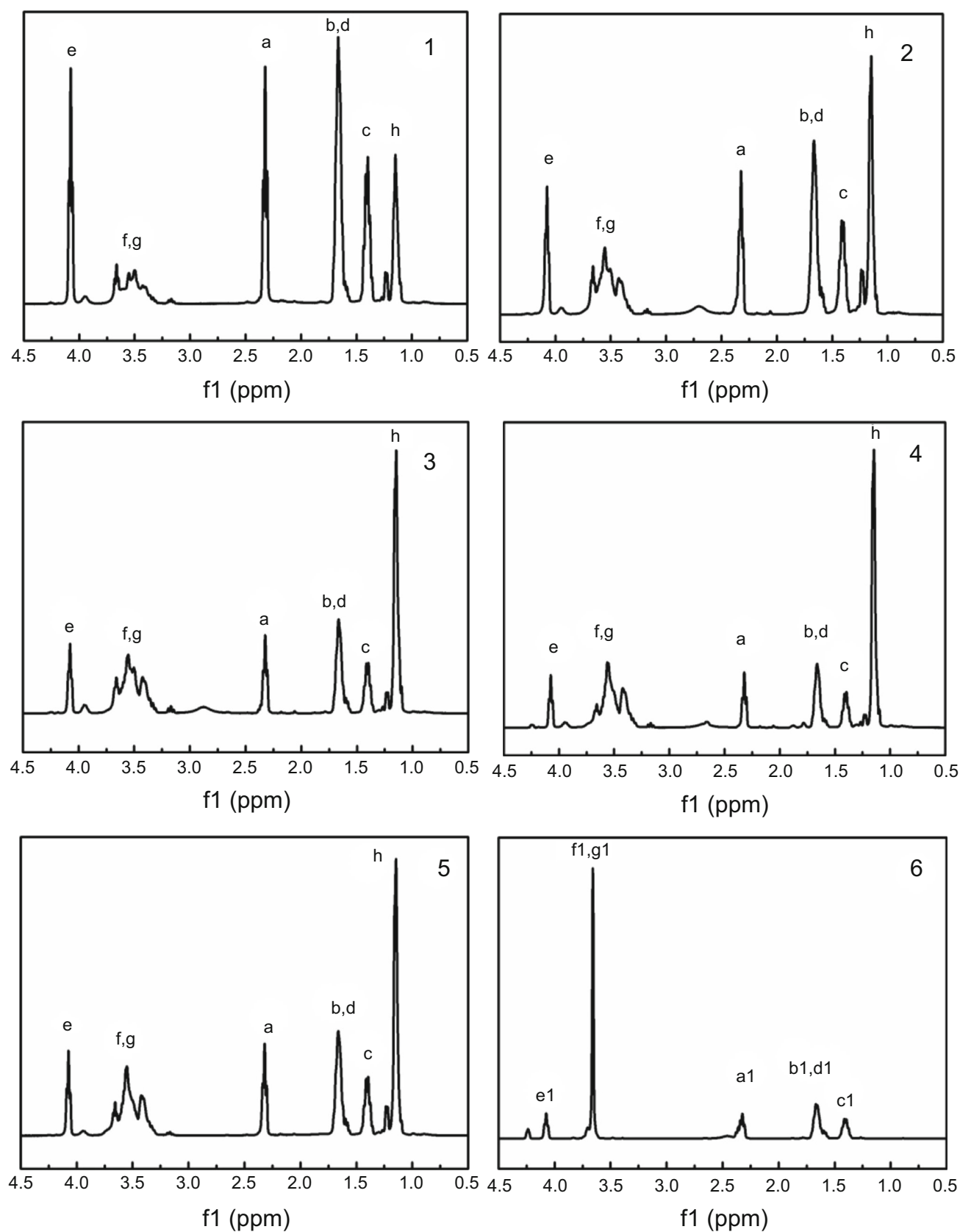


Fig. 1: ^1H NMR spectra of triblock polyols: (1) $\text{PPG}_{400}/\text{CL}_8$; (2) $\text{PPG}_{600}/\text{CL}_8$; (3) $\text{PPG}_{800}/\text{CL}_8$; (4) $\text{PPG}_{1000}/\text{CL}_8$; (5) $\text{PPG}_{1200}/\text{CL}_8$; (6) $\text{PEG}_{800}/\text{CL}_8$

(–NH–) is at 3321 cm^{-1} . The absorption bands at 2933 and 2865 cm^{-1} are due to asymmetric and symmetric stretching of $-\text{CH}_2-$. The stretching absorption of carbonyl group is at 1730 cm^{-1} . The characteristic bands of $\text{C}-\text{O}-\text{C}$ are at 1231 and 1064 cm^{-1} . The characteristic band of $\text{CO}-\text{NH}$ is at 1541 cm^{-1} . The

absence of the characteristic stretching band of $-\text{NCO}$ at 2275 cm^{-1} testified that $-\text{NCO}$ had all been cured.

The mechanical properties of PU films

The mechanical properties of PU films are shown in Table 2. The tensile strength and modulus of the PU films increased with decreasing the number average molecular weight of PPG_x . Additionally, the tensile strength and modulus of $\text{PPG}_{800}/\text{CL}_8$ -PU were higher than that of $\text{PEG}_{800}/\text{CL}_8$ -PU. It meant that the chain of polypropylene glycol could improve the tensile strength and modulus of PU film compared to the chain of polyethylene glycol. The elongation at break of $\text{PPG}_{800}/\text{CL}_8$ -PU was the highest of all the PPG_x/CL_8 -PU. Nevertheless, the elongation at break of $\text{PEG}_{800}/\text{CL}_8$ -PU was the highest of all prepared PU films. Therefore, it was considered that the mechanical properties of the prepared PU films could be controlled by varying the number average molecular weight of polypropylene glycol and the type of polyether.

Thermal properties of PU films

The crystallinity enthalpy of all PU films did not show in the DSC curve (Fig. 4). At the same time, the crystallite diffraction peaks (not shown) of all PU films by XRD were not obvious, which was in good agreement with the DSC curve of PU films. It was reported that PCL-PU having a PCL soft segment with a number average molecular weight of over 2000 could crystallize, and that its crystallinity increased with increasing molecular weight of PCL.²³ The DSC curve of $\text{PEG}_{800}/\text{CL}_8$ -PU showed an endotherm due to volatilization of water near this temperature (about 100°C) because of the extreme hydrophilicity of $\text{PEG}_{800}/\text{CL}_8$ -PU.²³

The Tgs of PU films are shown in Table 2. The Tg of PU film decreased with increasing the length of polypropylene glycol. PU film is composed of hard and soft segments. The soft segments determine flexibility and Tg of PU films. The Tgs of PU films decreased slightly with increasing soft segment content

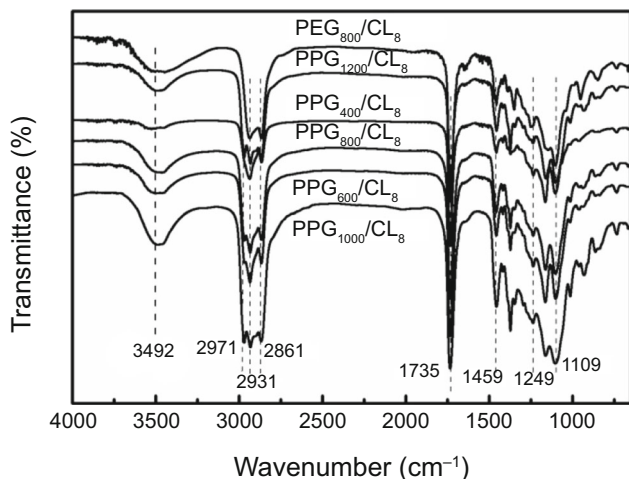


Fig. 2: The FTIR spectra of triblock polyols

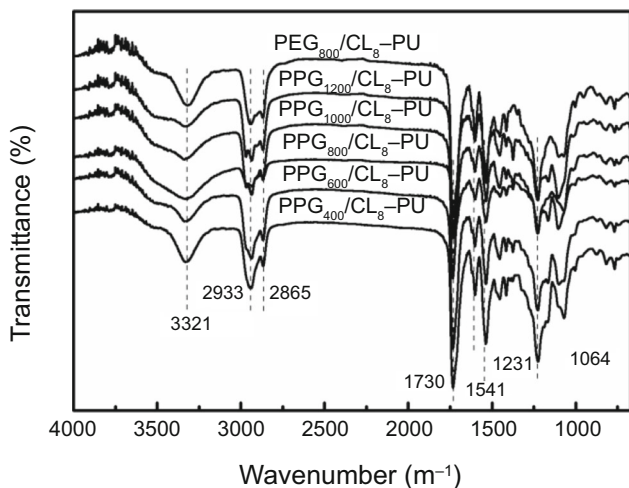


Fig. 3: The ATR-FTIR spectra of PU films

Table 2: The mechanical properties and Tgs of PU films

Samples	Tensile strength (MPa)	Tensile modulus (MPa)	Elongation-at break (%)	Tg (°C)
PPG ₄₀₀ /CL ₈ -PU	11.63 ± 1.62	70.2 ± 5.3	49.88 ± 3.62	– 29.43
PPG ₆₀₀ /CL ₈ -PU	8.92 ± 1.05	20.5 ± 3.1	56.78 ± 4.21	– 31.17
PPG ₈₀₀ /CL ₈ -PU	6.84 ± 0.85	11.6 ± 2.2	60.15 ± 4.52	– 33.28
PPG ₁₀₀₀ /CL ₈ -PU	4.62 ± 0.84	8.3 ± 1.1	55.42 ± 2.95	– 36.29
PPG ₁₂₀₀ /CL ₈ -PU	4.15 ± 0.52	7.8 ± 0.9	52.62 ± 2.75	– 38.08
PEG ₈₀₀ /CL ₈ -PU	4.75 ± 4.21	7.3 ± 0.8	65.82 ± 4.52	– 34.73

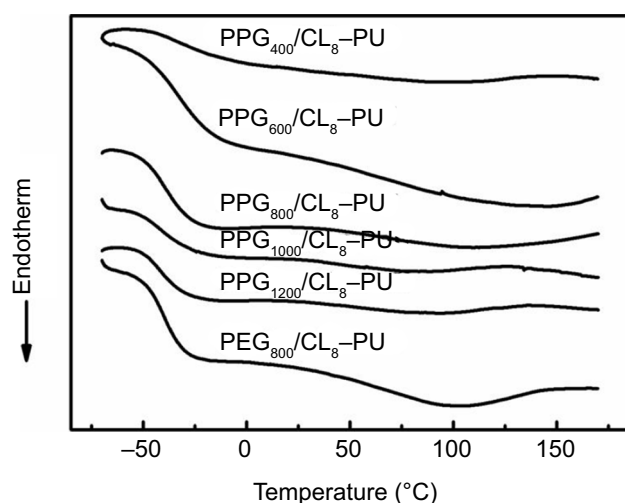


Fig. 4: The second heat DSC curves of PU films

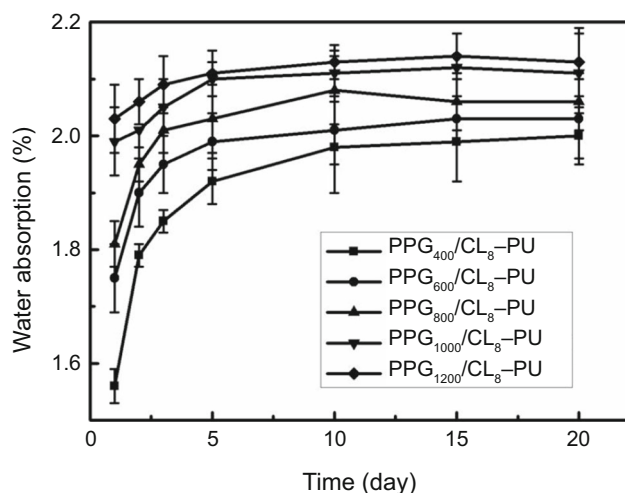


Fig. 5: The water absorption rates of PU films in artificial sea water

as a consequence of the increased mobility of the soft segments. Therefore, it was considered that the T_g of PU film could be controlled by varying the number average molecular weight of polypropylene glycol.

Water absorption rates of PU films

The properties of water absorption rates of PU films have received considerable attention due to the importance of these phenomena in the case of the hydrolytic degradation of PU films. The water absorption rates of PU films are shown in Fig. 5. It was found that the water absorption rates of PU films were greatly dependent on the number average molecular weight of PPG. The water absorption of PU films was constant for about 10 days. The water absorption of PPG₁₂₀₀/CL₈-PU was the highest of all the PU films, which indicated that the water absorption of PU films

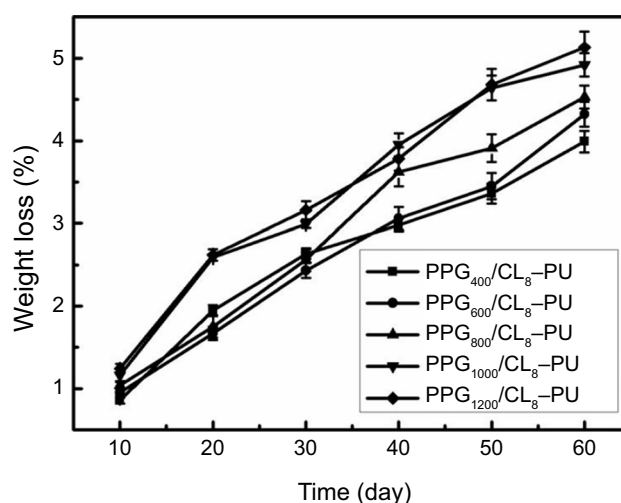


Fig. 6: The hydrolytic degradation rates of PU films in artificial sea water

improved with increasing soft segment content. Nevertheless, the water absorption of PEG₈₀₀/CL₈-PU was about 71.23% after 10 days (not shown in Fig. 5), which was much higher than those of PPG_x/CL₈-PU. Polyethylene glycol (PEG) is a nonionic and biocompatible material with extreme hydrophilicity. However, polypropylene glycol (PPG), derived from propylene glycol (PG) monomers with a methyl substituent at the ethylene glycol (EG) basic structure, is hydrophobic. Yet, the contrary results were apparently shown in this present study, indicating that the hydrophilic property of the polyether chain plays a more important role than the number average molecular weight of the polyether chain. The water absorption rates of triblock polyols depend on the hydrophilic property of the polyether chain.

Hydrolytic degradable properties of PU films

The hydrolytic degradable properties of PU films based on triblock polyols were studied to evaluate their potential applications as marine antifouling materials. Weight loss of PU films during degradation reflects the release of soluble oligomers from the PU films. Figure 6 shows the hydrolytic degradation rates of PU films in ASW. It was found that the hydrolytic degradation rates of PU films improved dramatically with increasing water absorption rates of PU films. Nevertheless, the hydrolytic degradation rate of PEG₈₀₀/CL₈-PU was about 31.24% after 30 days (not shown in Fig. 6), which was much higher than those of PPG_x/CL₈-PU. The hydrolytic degradation rate of PEG₈₀₀/CL₈-PU was too high, which could not be independently used as marine antifouling coatings. The results indicated that the hydrolytic degradation rates of PU films could be controlled by varying the water absorption rates of PU films.

Table 3: The mechanical properties of PU coatings

Samples	Adhesion strength (MPa)	Impact resistance (kg cm)
PPG ₄₀₀ /CL ₈ -PU	7.2 ± 0.4	50
PPG ₆₀₀ /CL ₈ -PU	5.3 ± 0.3	50
PPG ₈₀₀ /CL ₈ -PU	4.4 ± 0.3	50
PPG ₁₀₀₀ /CL ₈ -PU	3.4 ± 0.2	50
PPG ₁₂₀₀ /CL ₈ -PU	3.0 ± 0.2	50

Properties of PU coatings

The mechanical properties of PU coatings

The adhesion strength of PU coatings is shown in Table 3. The adhesion strength of PU coatings increased with decreasing the number average molecular weight of PPG_x. This is because the density of the hard segment decreases as the number average molecular weight of PPG_x increases, which reduces the polar interaction between the polymer and the substrate.²⁵ All the coatings exhibit excellent adhesion with a value higher than 3.0 MPa, which can meet the needs for marine antifouling.

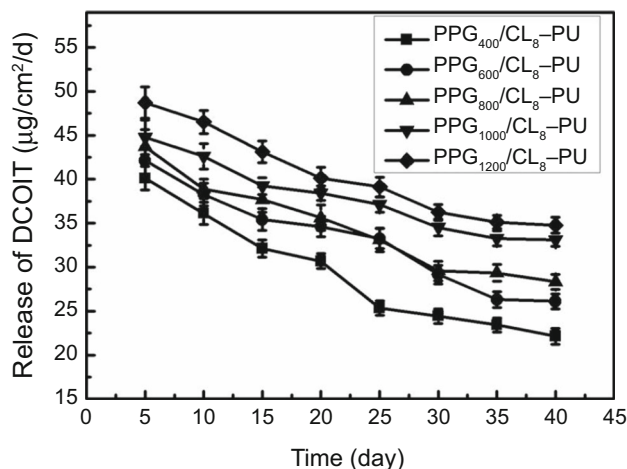
The impact resistance of PU coatings is shown in Table 3, the value of impact resistance of all PU coatings is 50 kg cm, and the high impact resistance exhibited by PU coatings indicates that they are less susceptible to fracture.

DCOIT release rates of PU coatings

For effective prevention of fouling, a constant and defined concentration of DCOIT is desirable at the surface of the coating. Therefore, it is necessary to determine the release rates of DCOIT immersed into seawater, which is controlled by the PU binder of coatings. The relationship between immersion time and DCOIT release rates of PU antifouling coatings is shown in Fig. 7, which displays the DCOIT release rates from PU coatings in ASW. It was found that DCOIT release rates of PU coatings improved dramatically with increasing hydrolytic degradation rates of PU films. The results suggested that DCOIT release rates from PU coatings were mainly determined by the hydrolytic degradation rates of PU film.

The erosion rate of the thickness of PU coatings

A month later, the mass loss was almost linear with time, indicating the degradation with constant rate. The erosion rate of the thickness of PPG_x/CL₈-PU coating is shown in Table 4. The erosion rate of the thickness of PPG₁₂₀₀/CL₈-PU coating was the highest (about 0.84 ± 0.04 μm/week) during the static immer-

**Fig. 7: The DCOIT release rates of PU antifouling coatings**

sion. It was found that the erosion rate of the thickness of PPG_x/CL₈-PU coating improved dramatically with increasing hydrolytic degradation rates of PU films.

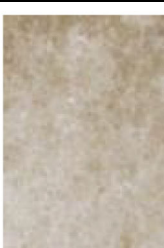
Assessment of antifouling performance

The marine antifouling performances of PU antifouling coatings were examined. The surfaces of steel panels coated with PU antifouling coatings were almost clean after immersion in sea water for 3 months, shown in Table 5. For PPG₄₀₀/CL₈-PU, PPG₆₀₀/CL₈-PU, and PPG₈₀₀/CL₈-PU coatings, they were fouled because of the lower self-polishing rate and the release rate of DCOIT. PPG₁₀₀₀/CL₈-PU and PPG₁₂₀₀/CL₈-PU coatings were almost clean after immersion in sea water for 9 months, shown in Table 5. These results indicated that PPG₁₀₀₀/CL₈-PU and PPG₁₂₀₀/CL₈-PU coatings had the properties of effective marine antifouling because a constant and defined concentration of DCOIT was desirable at the surfaces of the coatings. Therefore, it was demonstrated that the degradable polyurethanes based on triblock polyols composed of polypropylene glycol and ε-caprolactone could be used for marine antifouling.

Table 4: The erosion rate of the thickness of PU coatings

Erosion rate	PPG ₄₀₀ /CL ₈ -PU	PPG ₆₀₀ /CL ₈ -PU	PPG ₈₀₀ /CL ₈ -PU	PPG ₁₀₀₀ /CL ₈ -PU	PPG ₁₂₀₀ /CL ₈ -PU
μm/week	0.61 ± 0.02	0.72 ± 0.02	0.75 ± 0.03	0.81 ± 0.04	0.84 ± 0.04

Table 5: The typical images of panels with PU coatings for antifouling

Time	PPG ₄₀₀ /CL ₈ -PU	PPG ₆₀₀ /CL ₈ -PU	PPG ₈₀₀ /CL ₈ -PU	PPG ₁₀₀₀ /CL ₈ -PU	PPG ₁₂₀₀ /CL ₈ -PU
3 month					
6 month					
9 month					

Conclusions

Degradable polyurethane (PU) coatings have a reputation for high performance. The degradable triblock polyols composed of polyether (PPG_x or PEG₈₀₀) and CL were prepared. The degradable PU films were prepared by mixing triblock polyols and crosslinker agent (L75). The hydrolytic degradation and water absorption experiments of PU films demonstrated that the PU films could erode in artificial sea water, which were mainly controlled by the hydrophilic property of the polyether. The hydrolytic degradation rate of PPG₈₀₀/CL₈-PU was about 2.43% after 30 days. Nevertheless, the hydrolytic degradation rate of PEG₈₀₀/

CL₈-PU was about 31.24% after 30 days. Marine field test revealed that degradable polyurethane (PPG₁₀₀₀/CL₈-PU and PPG₁₂₀₀/CL₈-PU) had excellent efficiency in preventing biofouling for more than 9 months. We believe the degradable polyurethane based on triblock polyols composed of polypropylene glycol and ε-caprolactone is promising in the development of long-term marine antibiofouling coatings.

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